

Assignment 2 - Part 2: Multithreading with pthreads

Fall 2026

Due: Friday, Sep 25 11:59 pm

Coding Task 1: Convert the array sum program from Assignment 1 to parallel using pthreads. Use the starter code sum_T.c

Hint: The summation work should be divided equally among the threads.

Coding Task 2: Convert the matrix multiplication program from Assignment 1 to parallel using pthreads. Use the starter codes matrixT.h and matrixT.c

Hint: The threads divide the work by rows.

Task 3: Benchmark the time it took for both your sequential and parallel programs using a table as shown below:

Array Sum Problem (for array size 100,000,000)					
	Sequential	1 thread	2 threads	3 threads	4 threads
Time					
Matrix Multiplication Problem (for matrix size 1,000)					
	Sequential	1 thread	2 threads	3 threads	4 threads
Time					

Task 4: Explain your observations from Task 3 in detail. Did your solution require the use of synchronization primitives? Explain why synchronization was or was not necessary for the correctness of your implementation. Will this manual synchronization technique work? Why or why not?

```
if(lock == 0)
{
    lock = 1;
    // some critical section code
    lock = 0;
}
```

Deliverables:

- A zipped folder named A2_part_2 submitted to D2L dropbox that contains:
 - o Your source code files for both coding tasks 1 and 2.
 - o A pdf file for Task 3, Task 4, and a screenshot showing your codes work as expected.
 - o A README.txt file
- Available on Github inside the folder where you added me as a collaborator.

Total Points (50)

- Code runs and works as expected – Task 1: 10 points
- Code runs and works as expected – Task 2: 20 points
- Clear documentation – Task 3: 5 points
- Clear and detailed explanation – Task 4: 10 points
- Available on GitHub: 5 points